

PAPER_745: Crab Nebula MUGE — Expanding Supernova Remnant with Pulsar Wind

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Framework: Universal Quantum Field Superconductive Framework (UQFF)

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Abstract

The Crab Nebula (M1) is the remnant of SN 1054, powered by a central pulsar (PSR B0531+21) at the highest rotational energy loss in the known neutron star population. Its expanding ejecta ($r(t) = r_0 + v_r \cdot t$) creates a time-dependent gravitational environment unlike any static system. This paper derives the Crab Nebula MUGE incorporating: the expanding radius $r(t)$, the pulsar wind term F_{wind} , and the nebular magnetic energy M_{mag} . The UQFF framework captures both the mechanical expansion and the magnetospheric energy injection that power the synchrotron nebula.

1. Introduction

The Crab Nebula is a pulsar wind nebula (PWN) — one of the brightest objects in X-rays and gamma-rays. The central pulsar ($P = 33$ ms, $\dot{P} = 4.2 \times 10^{-13}$ s/s) injects $\sim 5 \times 10^{31}$ W of spin-down power into the nebulae through a relativistic wind. The expanding ejecta shell moves at $v_r \sim 1500$ km/s, with angular size growing observably over human timescales. This expanding geometry requires $r(t)$ to appear explicitly in the MUGE denominator, making the Crab the canonical example of a time-dependent gravitational radius.

Crab Nebula parameters: - Distance = 2.0 kpc - Age ≈ 970 yr (SN 1054) - r_0 (initial remnant radius) $\approx 3 \times 10^{15}$ m - $r_{current} \approx 5.6$ ly = 5.3×10^{16} m - $v_r = 1500$ km/s = 1.5×10^6 m/s - $M_{ejecta} \approx 4\text{--}5 M_\odot$ - $B_{nebula} \approx 10^{-4}$ T (filamentary magnetic field) - $L_{pulsar} = 5 \times 10^{31}$ W (spin-down luminosity) - $P_{spin} = 33$ ms

2. Crab Nebula MUGE

$$\begin{aligned} g_{Crab}(r,t) = & (G \cdot M) / r(t)^2 \cdot (1 + H(z) \cdot t) \cdot (1 - B(t) / B_{crit}) \\ & + (U_{g1} + U_{g2} + U_{g3} + U_{g4}) \\ & + U_i \\ & + (\Lambda \cdot c^2 / 3) \\ & + (\hbar / \sqrt{\Delta x \cdot \Delta p}) \cdot \int (\psi \cdot H \cdot \psi \, dV) \cdot (2\pi / t_{Hubble}) \\ & + \rho_{ejecta} \cdot V \cdot g \\ & + F_{wind} \\ & + M_{mag} \end{aligned}$$

[pulsar wind -- NEW]
[magnetic energy -- NEW]

3. Time-Dependent Radius $r(t)$

The expanding radius is the defining feature of the Crab MUGE:

$$r(t) = r_0 + v_r \cdot t$$

r_0 = initial radius at SN explosion (m)
 v_r = expansion velocity = 1.5×10^6 m/s
 t = time since SN 1054 (s)

For $t = 970 \text{ yr} = 3.06 \times 10^{10} \text{ s}$:

$$r(970 \text{ yr}) = 3 \times 10^{15} + 1.5 \times 10^6 \times 3.06 \times 10^{10}$$

$$r(970 \text{ yr}) \approx 4.6 \times 10^{16} \text{ m} \approx 4.9 \text{ ly}$$

Gravitational deceleration:

$$g_{\text{grav}}(t) = G \cdot M_{\text{ejecta}} / r(t)^2$$

$$g_{\text{grav}}(970 \text{ yr}) = 6.674 \times 10^{-11} \times (4 \times 4 \times 1.989 \times 10^{30}) / (4.6 \times 10^{16})^2$$

$$g_{\text{grav}} \approx 5 \times 10^{-22} \text{ m/s}^2 \quad (\text{extremely weak} - \text{expansion dominates})$$

4. F_{wind} — Pulsar Wind Term

The pulsar wind carries relativistic particles outward, creating an inward reaction force equivalent to:

$$F_{\text{wind}} = L_{\text{pulsar}} / (4 \cdot \pi \cdot r(t)^2 \cdot c \cdot M_{\text{ejecta}})$$

$L_{\text{pulsar}} = 5 \times 10^{31} \text{ W}$
 $r(t)$ = current nebula radius
 $c = 3 \times 10^8 \text{ m/s}$

$$F_{\text{wind}} = 5 \times 10^{31} / (4 \cdot \pi \cdot (4.6 \times 10^{16})^2 \cdot 3 \times 10^8 \cdot 4 \times 1.989 \times 10^{30})$$

$$F_{\text{wind}} \approx 4.8 \times 10^{-30} \text{ m/s}^2 \quad (\text{small but physically significant over 970 yr})$$

Cumulative momentum injection over nebula lifetime:

$$\Delta p_{\text{wind}} = L_{\text{pulsar}} \cdot t / c \approx 5 \times 10^{31} \times 3 \times 10^{10} / 3 \times 10^8 \approx 5 \times 10^{33} \text{ kg} \cdot \text{m/s}$$

5. M_{mag} — Magnetic Energy Term

The nebular magnetic field stores and dissipates energy, affecting dynamics:

$$M_{\text{mag}} = B^2 / (2 \cdot \mu_0 \cdot r(t) \cdot \rho_{\text{ejecta}}) \quad [\text{magnetic pressure divided by density}]$$

$B = 10^{-4} \text{ T}$ (filamentary field)
 $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$
 $\rho_{\text{ejecta}} \approx 10^{-22} \text{ kg/m}^3$ (current)

$$M_{\text{mag}} = (10^{-4})^2 / (2 \cdot 4\pi \times 10^{-7} \cdot 4.6 \times 10^{16} \cdot 10^{-22})$$

$$M_{\text{mag}} \approx 2.7 \times 10^{-17} \text{ m/s}^2$$

This is the dominant non-Newtonian term for the Crab, controlling the synchrotron brightness evolution.

6. UQFF Gravity Components

U_{g1} : Pulsar magnetic dipole
 $\mu_{\text{dipole_PSR}} \approx 3.8 \times 10^{30} \text{ J/T}$ (from P , $\dot{P}$)

$B_{\text{PSR}} = 3.8 \times 10^8 \text{ T}$ (surface field)
 U_{g1} contributes at pulsar vicinity only

U_{g2} : Superconductive aether field threading nebula
 B_{super} aligned with pulsar rotation axis
 $U_{\text{g2}} \approx 10^5 \text{ J/m}^3$ (strong near pulsar wind shock)

U_{g3} : External galactic field at 2 kpc
 $U_{\text{g3}} = G \cdot M_{\text{gal}} / r_{\text{gal}}^2$

U_{g4} : Galactic center contribution
 $k_4 \cdot \rho_{\text{vac}} [\text{SCm}] \cdot (M_{\text{bh}} / d_{\text{g}}) \cdot e^{(-\alpha t)} \cdot \cos(\pi \cdot t_n)$

7. Temporal Evolution Summary

Time (yr)	$r(t)$ (ly)	g_{grav} (m/s ²)	M_{mag} (m/s ²)
0	0.3	2×10^{-14}	10^{-16}
100	1.6	5×10^{-21}	10^{-17}
970	4.9	5×10^{-22}	2.7×10^{-17}
10,000	~50	$\sim 5 \times 10^{-27}$	$\sim 10^{-19}$

The magnetic term M_{mag} remains important relative to g_{grav} throughout nebula evolution.

8. Conclusion

The Crab Nebula MUGE demonstrates that time-dependent radius $r(t)$ is essential for accurately modeling supernova remnant gravity. The F_{wind} pulsar injection and M_{mag} magnetic energy terms together dominate over classical Newtonian gravity within the nebula. The UQFF successfully models the transition from SN ejecta to mature PWN through its modular environmental forcing framework.

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